

Plate Reader Linearity Methods

A synthetic readout model for the SpectraNova SN-4 optical workflow.

This methods note focuses on the instrument contribution to a fictional planning workflow. It does not claim that a plate-reader fit proves a biological effect, sample concentration, or assay suitability beyond its stated synthetic review context.

Readout model

Model component	Synthetic value	Method consequence
Instrument	SpectraNova SN-4	retain serial context in review packet
Calibration slope	0.998	compare against expected proportional response
Fit quality	R ² 0.9992	inspect residuals before relying on fit
Blank response	0.02 AU	apply blank check before interpretation
Read wavelength	450 nm planning channel	keep wavelength context with export

Residual-aware reading

A high fit statistic does not replace inspection of standards, blank behavior, or the chosen wavelength. The synthetic workflow treats the 0.02 AU blank as an operational check: it is a prompt to review baseline handling, not a measurement of a real material.

Linearity review implications

From staged standards to a bounded instrument statement.

Question	Evidence artifact	Permitted interpretation
Are standards ordered?	staged standard series	readout sequence is reviewable
Does proportionality hold?	slope 0.998 and residual note	fit is consistent with synthetic planning model
Is the baseline controlled?	0.02 AU blank check	background was considered before summary
What wavelength was used?	450 nm planning channel	context travels with export
What cannot be inferred?	no specimen or patient data	no biological conclusion

Conclusion

The SpectraNova SN-4 record supports only a narrow operational conclusion: the fictional instrument context, standards, and blank check are explicit enough for an internal reviewer to discuss readout quality. It does not validate an assay outside this synthetic packet.